Maximal divergence of the leximin and leximax core selections in exact games

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Abstract

A singleton-normalized exact transferable-utility game carries two canonical egalitarian core selections: the leximin allocation Lmin and the leximax-min allocation Lmax. They coincide on every exact game with at most four players. We measure their separation by $\rho(v) = \|\operatorname{Lmin}(v) - \operatorname{Lmax}(v)\|_1 / v(N)$, which never exceeds 2, and show that 2 is the exact supremum. The witnesses form an explicit two-parameter family $G(p,k)$ of two-generator lower-envelope games with $\rho(G(p,k)) = 2(p(k-1)-1)/(k(p+1))$ and the deficit identity $2 - \rho(G(p,k)) = 2n/(k(p+1))$, $n = p+1+k$, so that $s(n) \geq 2 - 8n/(n^2 - (n \bmod 2))$ for every $n \geq 4$; the ratio reaches 1 at eight players and exceeds it from nine on. The mechanism is a single family of tight $(p+1)$-player caps that both selections saturate: leximin reads a cap as a ceiling and piles the residual onto one player, leximax-min reads it as a budget and spreads the residual over half the players, so the two supports become asymptotically disjoint. The concentrating selection is the leximin one. On the two-generator class we prove a dual description of the core, $\operatorname{Core}(v) = \{x \geq 0 : x(N) = T,\ x(S) \leq \max(g(S), h(S))\}$, deduce a one-coalition upper bound on the leximin floor, and exhibit the smallest game in a complete enumeration at which the mirror bound for the leximax-min ceiling fails. At $n = 5$ the family formula is not sharp: a five-player game analysed by hand attains $1/2$, and a configuration-complete exact finite check — run twice, through two pipelines sharing no machinery — proves it extremal: $s(5) = 1/2$, with the sharper bound $1/3$ when the leximin strata form the pattern $(4,1)$, both attained. We also prove that at five players *both* egalitarian levels are certified by single coalitions — the property whose failure from six players on accompanies the climb of $s(n)$. At six players a direct exactness-exchange finite proof gives the matching upper bound $s(6) = 2/3$: fifteen sign-count relaxations are certified infeasible by independently checked exact proof objects. We also close one natural route to a matching upper bound, by a seven-player game in which smoothing a generator toward uniformity collapses ρ from $1/5$ to 0.

1 Introduction

Let (N, v) be a transferable-utility game on $N = \{1, \ldots, n\}$ with

$$\operatorname{Core}(v) = \{x \in \mathbb{R}^N : x(N) = v(N),\ x(S) \geq v(S) \text{ for all } S \subseteq N\},$$

where $x(S) = \sum_{i \in S} x_i$. The game is *exact* if every coalition inequality is attained by some core allocation, and *singleton-normalized* if $v(\{i\}) = 0$ for every i; on that domain every core allocation is nonnegative. Exact games are the natural home for egalitarian core selections, and two such selections are canonical. The *leximin* allocation $\operatorname{Lmin}(v)$ maximizes the increasing rearrangement of x lexicographically over the core; the *leximax-min* allocation $\operatorname{Lmax}(v)$ minimizes the decreasing rearrangement lexicographically over the core. Both are single-valued on a nonempty compact

convex set. The first raises the poorest as far as possible, the second lowers the richest as far as possible.

On small games the two agree. The companion paper (Flanagan, 2026) proves that every singleton-normalized exact game on at most four players is an exact partition game at Lmin; by the established implication from exact partition to Lorenz domination (Llerena and Mauri, 2017), Lmin then Lorenz-dominates every core allocation, and a Lorenz-greatest core point is simultaneously leximin- and leximax-optimal[1], so $\mathrm{Lmin}(v) = \mathrm{Lmax}(v)$ for every singleton-normalized exact game with $n \le 4$. Our complete enumerations at $n \le 4$ reconfirm this rigidity as a control (Section 5). The adjacent large-core chain — exact games on at most four players have large cores (Biswas et al., 1999), and on large-core games the procedural egalitarian solution coincides with Lmax (Dietzenbacher, 2021; Arin et al., 2003) — pins only the Lmax side: it does not deliver the coincidence, and indeed a five-player exact game with a large core has $\mathrm{Lmin} \neq \mathrm{Lmax}$: the lower envelope of $(2, 0, 12, 6, 0)$, $(0, 0, 0, 0, 20)$ and $(0, 2, 10, 4, 4)$, with $\|\mathrm{Lmin} - \mathrm{Lmax}\|_1 = v(N)/5$ (verified exactly; archived with the repository). The companion's lower-envelope realization technique is used again here.

The pair itself has recent precedent. Vazirani (2024) and Gangam et al. (2024) study leximin and leximax core imputations of assignment, flow and matching games, observe on examples that the two selections differ, and describe their opposition in exactly the direction this paper quantifies; the leximax selection appears earlier as the leximax solution of neighbour games (Klijn et al., 2003) and as the lexmax solution in the egalitarian-set literature (Llerena and Mauri, 2016), besides its large-core characterization (Arin et al., 2003). The candidate contribution here is the quantitative worst case: the exact domain, the supremum 2 with its support-disjointness characterization, the closed-form extremal family, and the direction result that the *concentrating* selection is the leximin one. A targeted review of the principal egalitarian-core, antiduality, exact-partition, large-core, and exact-cone sources found no prior fixed-dimensional separation bound (Arin et al., 2008; Dietzenbacher and Yanovskaya, 2020; Studený and Kratochvíl, 2022); this negative search is not a novelty proof, and the claim remains subject to specialist review.

The question of this paper is what happens after the rigidity breaks. Write

$$\rho(v) = \frac{\|\mathrm{Lmin}(v) - \mathrm{Lmax}(v)\|_1}{v(N)}, \qquad s(n) = \sup\{\rho(v) : \ v \text{ exact on } n \text{ players}, \ v(N) > 0\},$$

the supremum running over singleton-normalized games. Since $\mathrm{Lmin}(v)$ and $\mathrm{Lmax}(v)$ are nonnegative with the same total $v(N)$, we have $\rho \le 2$ always, and $\rho = 2$ would mean disjoint supports. The rigidity says $s(n) = 0$ for $n \le 4$. How large can $s(n)$ get?

Results. The answer is: as large as arithmetically possible.

- $\sup_n s(n) = 2$, and at fixed n, $s(n) \ge 2 - 8n/(n^2 - (n \bmod 2))$, which is $2 - 8/n$ for even n (Theorem 17); in particular $s(n) \ge 1$ from $n = 8$ and $s(n) > 1$ from $n = 9$; the bound is the full exact-class value at $n = 6$, and is attained on the two-generator class at $n = 7, 8$ by complete enumeration.
- The witnesses form a two-parameter family $G(p, k)$ whose two allocations are computed in closed form (Theorem 13), with the deficit identity $2 - \rho(G(p, k)) = 2n/(k(p + 1))$ (Corollary 15).

[1]With equal totals, Lorenz domination gives $\sum_{i \le k} x_i^{\uparrow} \ge \sum_{i \le k} y_i^{\uparrow}$ for every k; induction on k yields $x^{\uparrow} \ge_{\mathrm{lex}} y^{\uparrow}$, and complementing partial sums gives the leximax-min half. See Dutta and Ray (1989) for the Lorenz-dominant core element on convex games.

- On the two-generator class the core has a dual description as a system of upper caps (Theorem 6), from which the leximin floor inherits a one-coalition certificate (Theorem 7); the mirror statement for the leximax-min ceiling is false, and we give the smallest counterexample (Observation 10).
- $n = 5$ is anomalous: the family formula gives $1/3$, a game analysed by hand attains $1/2$ (Theorem 21), and the value is the supremum — $s(5) = 1/2$, with $\rho \leq 1/3$ in the $(4, 1)$ leximin regime (Theorems 33 and 32, by configuration-complete finite check, twice over) — while the $n = 5$ extremal class is qualitatively unlike the $n \geq 6$ one.
- At $n = 6$ the family lower bound is sharp: $s(6) = 2/3$ (Theorem 34). The upper bound is a table-free exact finite proof using singleton exactness witnesses and lexicographic exchange blockers; all fifteen sign-count cases have independently checked CPC proof objects.

The mechanism, in one sentence. In $G(p, k)$ a single family of tight caps forces $p+1$ players to share only p; leximin reads each cap as a ceiling, so the floor sticks at $p/(p+1)$ and the entire residual piles onto one player, while leximax-min reads the same cap as a budget, loading half the players up to $p(k-1)/k$ and thereby squeezing the capped players down to $1/k$ — so the two supports become asymptotically disjoint and $\rho \to 2$. It is worth stating the direction explicitly, because it inverts the usual reading: the *concentrating* selection is the leximin one. In $G(p, k)$ the leximin allocation gives one player a share $p(k-1)/((p+1)k) \to 1$ of $v(N)$ while everyone else sits just below 1; it is nevertheless leximin-optimal, because nothing better is feasible for the poorest. Both selections are driven into extreme shapes by the same constraint, in opposite directions.

What is not proved. An upper bound matching Theorem 17 at fixed n is open, and Section 6 shows that one natural route to it is closed. Section 7 separates the proved statements from the enumerated and the conjectured ones, and lists the open problems.

2 Preliminaries

Throughout, $N = \{1, \ldots, n\}$, all games are singleton-normalized, and $T := v(N) > 0$. Vectors are indexed by N, and $\mathrm{supp}(x) = \{i : x_i \neq 0\}$.

2.1 Exact games as lower envelopes

The following generation fact is the only construction used in this paper. It is standard; see Csóka et al. (2011) and Studený and Kratochvíl (2022) for the surrounding theory of the exact cone, and Flanagan (2026) for the same statement in the companion.

Lemma 1 (lower-envelope generation)**.** *Let $g_1, \ldots, g_r \in \mathbb{R}^N_+$ have a common total $g_1(N) = \cdots = g_r(N) = T$, and set $v(S) = \min_j g_j(S)$. Then v is exact, $v(N) = T$, every g_j lies in* Core(v)*, and v is singleton-normalized as soon as every coordinate vanishes in some g_j. Conversely, every exact singleton-normalized game is of this form: one may take $g_1, \ldots, g_r$ to be the vertices of* Core(v)*.*

Proof. Each g_j satisfies $g_j(S) \geq v(S)$ by definition of the minimum and $g_j(N) = T = v(N)$, so $g_j \in \mathrm{Core}(v)$; and the inequality at S is attained by whichever g_j realizes the minimum, so v is exact. The singleton statement is immediate from $v(\{i\}) = \min_j (g_j)_i$. Conversely, if v is exact then $v(S) = \min_{x \in \mathrm{Core}(v)} x(S)$ for every S, and the minimum of a linear functional over the polytope Core(v) is attained at a vertex; the vertices are nonnegative because v is singleton-normalized, and all have total T. □

We call a game generated as in Lemma 1 by r allocations an *r-generator* game. Part of what follows is specific to $r = 2$.

2.2 The two selections

For $x \in \mathbb{R}^N_+$ let $x^\uparrow$ be its increasing and $x^\downarrow$ its decreasing rearrangement.

Definition 2. $\mathrm{Lmin}(v)$ *is the allocation in* $\mathrm{Core}(v)$ *whose increasing rearrangement is lexicographically largest;* $\mathrm{Lmax}(v)$ *is the allocation in* $\mathrm{Core}(v)$ *whose decreasing rearrangement is lexicographically smallest.*

Both exist and are unique on a nonempty compact convex set: existence is compactness applied stage by stage, and uniqueness follows by comparing two putative optima with their midpoint, which is feasible by convexity and strictly better at the first stage where they differ. We write $\mu(v) = \min_i \mathrm{Lmin}(v)_i$ for the *leximin floor* and $M(v) = \max_i \mathrm{Lmax}(v)_i$ for the *leximax-min ceiling*; equivalently $\mu = \max_{x \in \mathrm{Core}} \min_i x_i$ and $M = \min_{x \in \mathrm{Core}} \max_i x_i$.

2.3 Normalization and the ratio

Two conventions have to be stated honestly, because neither selection is translation-equivariant: both are applied to the raw payoff vector x, not to $x - (v(\{i\}))_{i \in N}$, and on a game that is not singleton-normalized these are different objects — the leximax-min allocation in particular changes under the shift. We therefore fix the domain once and for all: all games are singleton-normalized, and both selections are computed on the normalized game. No claim below asserts invariance of Lmin or Lmax under adding singleton values.

Scaling, by contrast, is harmless: $\mathrm{Core}(cv) = c\,\mathrm{Core}(v)$ for $c > 0$ and both selections are positively homogeneous, so $\mathrm{Lmin}(cv) = c\,\mathrm{Lmin}(v)$ and $\mathrm{Lmax}(cv) = c\,\mathrm{Lmax}(v)$. Hence $\rho(v) = \|\,\mathrm{Lmin}(v) - \mathrm{Lmax}(v)\|_1 / v(N)$ is scale-invariant, which is what makes it comparable across games of different size. We record the trivial ceiling and the form of the deficit.

Lemma 3 (ceiling and overlap)**.** *For any singleton-normalized exact v with $T = v(N) > 0$,*

$$\rho(v) = 2\Big(1 - \frac{1}{T}\sum_{i \in N} \min\big(\mathrm{Lmin}(v)_i, \mathrm{Lmax}(v)_i\big)\Big) \;\le\; 2,$$

with $\rho(v) = 2$ if and only if $\mathrm{Lmin}(v)$ *and* $\mathrm{Lmax}(v)$ *have disjoint supports.*

Proof. For nonnegative x, y with $x(N) = y(N) = T$ one has $|x_i - y_i| = x_i + y_i - 2\min(x_i, y_i)$, and summing gives $\|x - y\|_1 = 2T - 2\sum_i \min(x_i, y_i)$. The overlap term is nonnegative and vanishes exactly when no coordinate is positive in both. □

So $\rho \to 2$ is precisely asymptotic support disjointness of the two egalitarian selections, and the whole content of the paper is that this can be forced.

3 The two-generator class

Throughout this section v is the lower envelope of exactly two nonnegative allocations g, h with common total T, so that $v(S) = \min(g(S), h(S))$ for every S (including $S = N$, where both sides equal T).

Lemma 4 (support disjointness)**.** *If v is singleton-normalized then $A := \mathrm{supp}(g)$ and $B := \mathrm{supp}(h)$ are disjoint, and every player outside $A \cup B$ is null, receiving 0 in every core allocation. Deleting null players changes neither* Lmin, Lmax, *nor T, so we may and do assume $A \cup B = N$ and $A \cap B = \emptyset$.*

Proof. $v(\{i\}) = \min(g_i, h_i) = 0$ forces $g_i = 0$ or $h_i = 0$, which is exactly $A \cap B = \emptyset$. If $i \notin A \cup B$ then $g_i = h_i = 0$, so $v(N \setminus \{i\}) = \min\big(g(N \setminus \{i\}), h(N \setminus \{i\})\big) = T$; any $x \in \mathrm{Core}(v)$ has $x(N \setminus \{i\}) \geq T$ and $x(N) = T$ with $x \geq 0$, hence $x_i = 0$. Both selections assign a null player 0, so deletion is harmless. □

Definition 5. *For $S \subseteq N$ put $f(S) := \max\big(g(S), h(S)\big) = \max\big(g(S \cap A), h(S \cap B)\big)$.*

3.1 A dual description of the core

Theorem 6 (two-generator core duality)**.**

$$\mathrm{Core}(v) = \big\{x \geq 0 : \ x(N) = T \text{ and } x(S) \leq f(S) \text{ for all } S \subseteq N\big\}.$$

Proof. For x with $x(N) = T$ and any S, the inequality $x(S) \geq v(S)$ is equivalent to $x(N \setminus S) \leq T - v(S)$. It therefore suffices to prove

$$T - v(S) = f(N \setminus S) \qquad \text{for every } S \subseteq N. \tag{1}$$

Since g and h are modular with $g(N) = h(N) = T$,

$$T - v(S) = T - \min\big(g(S), h(S)\big) = \max\big(T - g(S),\, T - h(S)\big) = \max\big(g(N \setminus S),\, h(N \setminus S)\big) = f(N \setminus S).$$

Replacing S by $N \setminus S$ in (1) turns the family $\{x(S) \geq v(S)\}$ into the family $\{x(S) \leq f(S)\}$. Finally $x \geq 0$ is exactly the family of singleton inequalities $x(\{i\}) \geq v(\{i\}) = 0$. □

Identity (1) uses only modularity and the common total; the disjointness of Lemma 4 is not needed for it, and enters only to normalize the presentation of f. Note also that $S = N$ gives $f(N) = T$, so the cap at N is implied by efficiency, and $S = \emptyset$ gives $f(\emptyset) = 0$.

In fact the duality is fully general: for *any* singleton-normalized game, setting $f(S) := T - v(N \setminus S)$, the same complementation involution gives $\mathrm{Core}(v) = \{x \geq 0 : x(N) = T,\ x(S) \leq f(S)\ \forall S\}$. The two-generator content of Theorem 6 is the computation $f = \max(g, h)$. The general form is used in Section 5.4.

3.2 The leximin floor

Theorem 7 (leximin floor: the easy direction)**.** *Let*

$$\lambda := \min_{\emptyset \neq S \subseteq N} \frac{f(S)}{|S|} = \min_{(A', B') \neq (\emptyset, \emptyset)} \frac{\max\big(g(A'), h(B')\big)}{|A'| + |B'|},$$

the second minimum running over $A' \subseteq A$, $B' \subseteq B$. Then $\mu \leq \lambda$.

Proof. $\mathrm{Lmin}(v)$ is a core allocation, so by Theorem 6, for every nonempty S, $|S|\,\mu \leq \mathrm{Lmin}(v)(S) \leq f(S)$, i.e. $\mu \leq f(S)/|S|$. Take the minimum over S. □

The bound is a *one-coalition certificate*: a single tightest cap already pins the floor from above. Its natural habitat is polymatroid theory: when f is monotone submodular, the polytope of Theorem 6 is a polymatroid base polytope, the leximin floor *equals* λ, and the decreasingly-minimal and increasingly-maximal points coincide — the theory of lexicographically optimal bases (Fujishige, 1980), where $\rho = 0$ identically. The two-generator $f = \max(g, h)$ is not submodular in general; the conjecture below asks whether the floor identity survives the loss of submodularity, and the rest of the paper measures exactly how far the dec-min/inc-max coincidence fails. Empirically the bound is exact.

Conjecture 8 (leximin floor formula)**.** *$\mu = \lambda$ for every singleton-normalized two-generator exact game.*

Evidence and reduction. Conjecture 8 holds in 586 of 586 random two-generator games at $n = 4, 5, 6, 7$ (sample sizes 120, 218, 165, 83), with all arithmetic exact. What remains to prove is a clean linear-programming statement. Equality means that the system

$$y \geq 0, \qquad y(N) = T - n\lambda, \qquad y(S) \leq f'(S) := f(S) - \lambda|S|$$

is feasible; indeed, given such y, the allocation $x = y + \lambda\mathbf{1}$ satisfies $x(N) = T$, $x \geq \lambda\mathbf{1} \geq 0$ and $x(S) \leq f(S)$, hence lies in the core with floor at least λ. Note $f' \geq 0$ by definition of λ. By linear-programming duality,

$$\max\{y(N) :\ y \geq 0,\ y(S) \leq f'(S)\ \forall S\} = \min\Big\{\textstyle\sum_S w_S f'(S) :\ w \geq 0,\ \sum_{S \ni i} w_S \geq 1\ \forall i\Big\},$$

the minimum running over fractional covers. Taking $w = \mathbf{1}_{\{S=N\}}$ shows the minimum is at most $f'(N) = T - n\lambda$, so Conjecture 8 is exactly the assertion

$$\sum_S w_S f'(S) \ \geq\ f'(N) \qquad \text{for every fractional cover } w. \tag{2}$$

Proposition 9 (balanced covers)**.** *Inequality (2) holds for every exact (balanced) cover, i.e. whenever $\sum_{S \ni i} w_S = 1$ for all i.*

Proof. Balancedness gives $\sum_S w_S|S| = \sum_i \sum_{S \ni i} w_S = n$, so, using $f \geq g$ and the modularity of g,

$$\sum_S w_S f'(S) = \sum_S w_S f(S) - \lambda n \ \geq\ \sum_S w_S g(S) - \lambda n = \sum_i g_i \sum_{S \ni i} w_S - \lambda n = g(N) - \lambda n = f'(N),$$

the last step because $g(N) = T$ and $f'(N) = T - \lambda n$. The same computation works with h in place of g. □

What is missing is the case of covers that strictly over-cover some players. Writing $e_i := \sum_{S \ni i} w_S - 1 \geq 0$, the same computation gives

$$\sum_S w_S f'(S) \ \geq\ f'(N) + \max\Big(\sum_{i \in A} g_i e_i,\ \sum_{i \in B} h_i e_i\Big) - \lambda \sum_{i \in N} e_i,$$

and since $\lambda \leq f(\{i\})$, which equals g_i for $i \in A$ and h_i for $i \in B$, each inner sum dominates λ times its *own* block's excess but not the other block's. Closing that gap, or constructing the attaining core allocation directly, is the remaining work.

3.3 The leximax-min ceiling is not dual

The mirror-image statement is an inequality but not an identity, and this is the first place where the two selections part company.

Observation 10 (M-side asymmetry)**.** *For every core allocation, and in particular for* $\mathrm{Lmax}(v)$*,*

$$M \;\geq\; \max_{\substack{P\subseteq A,\ Q\subseteq B\\ (P,Q)\neq(\emptyset,\emptyset)}} \frac{\min\big(g(P),h(Q)\big)}{|P|+|Q|}. \tag{3}$$

Inequality (3) is not an identity: it fails strictly already at $n=6$*.*

Proof of the inequality. Let $x\in\mathrm{Core}(v)$ and put $S=P\cup Q$. Then $g(S)=g(P)$ and $h(S)=h(Q)$ because g vanishes on B and h on A, so

$$\max_i x_i \;\geq\; \frac{x(S)}{|S|} \;\geq\; \frac{v(S)}{|S|} = \frac{\min\big(g(P),h(Q)\big)}{|P|+|Q|}.$$

Apply this to $x=\mathrm{Lmax}(v)$. Equivalently, (3) is Theorem 6 read on complements. □

How to read (3). The maximum runs over pairs (P,Q) other than $(\emptyset,\emptyset)$, with the convention $g(\emptyset)=h(\emptyset)=0$. There is no infinity convention: a pair with an empty part contributes $\min(\cdot,0)=0$ and is simply inactive, so only pairs with *both* parts nonempty can bind. This is visible in the proof, where an empty part makes $v(S)$ the minimum of something and 0. Reading an empty part as $+\infty$ instead would reduce (3) to a maximum over singletons and give a different — and wrong — value; the misreading is easy enough that we spell the convention out.

The smallest counterexample. Scanning the complete two-generator enumeration in increasing T, no counterexample to equality in (3) exists at $n\leq 5$ for any T, nor at $n=6$ for $T\leq 4$. At $n=6$, $T=5$:

$$g=(3,2,0,0,0,0), \qquad h=(0,0,2,1,1,1),$$
$$\mathrm{Lmin}(v)=\Big(1,\tfrac{2}{3},\tfrac{4}{3},\tfrac{2}{3},\tfrac{2}{3},\tfrac{2}{3}\Big), \qquad \mathrm{Lmax}(v)=\Big(\tfrac{6}{5},\tfrac{4}{5},\tfrac{6}{5},\tfrac{3}{5},\tfrac{3}{5},\tfrac{3}{5}\Big),$$

so $M=6/5$, whereas the right-hand side of (3) equals 1, attained for instance at $P=\{1\},Q=\{3\}$ with $\min(3,2)/2=1$. The gap is $1/5$. One further counterexample class occurs at $n=6$, $T=5$: $g=(3,1,1)$, $h=(2,2,1)$ on the respective supports (its block-swapped relabelling is the same game up to permutation), again with $M=6/5$ against a bound of 1. Over the same random samples used for Conjecture 8, (3) is tight in 120/120 games at $n=4$, 218/218 at $n=5$, 160/165 at $n=6$ and 74/83 at $n=7$.

Remark 11 (diagnosis)**.** *Theorem 7 bounds the floor using only the upper caps* $x(S)\leq f(S)$*, and a single tightest cap can be saturated on its own, which is why one coalition certifies the optimum. Bounding the ceiling instead requires the lower bounds* $x(N\setminus S)\geq T-f(S)$*, and* minimizing *a maximum must respect several of them simultaneously; no single one certifies the optimum. This is a genuine asymmetry between the two selections, and it is the reason* Lmax *is the harder object throughout.*

4 The master family

Definition 12. *For integers $p \geq 1$ and $k \geq 2$ put $n = p+1+k$ and $T = pk$, and let $G(p,k)$ be the lower envelope of*

$$g = \big(p(k-1),\ \underbrace{1,\dots,1}_{p},\ \underbrace{0,\dots,0}_{k}\big), \qquad h = \big(\underbrace{0,\dots,0}_{p+1},\ \underbrace{p,\dots,p}_{k}\big).$$

We index the players $0,1,\dots,p+k$ and call player 0 the head*, $O := \{1,\dots,p\}$ the* ones*, and $B := \{p+1,\dots,p+k\}$; so $A = \{0\} \cup O$ has $p+1$ elements and $|B| = k$.*

Both generators are nonnegative with total $p(k-1)+p = pk = T$, their supports are disjoint, and every coordinate vanishes in one of them; by Lemma 1, $G(p,k)$ is exact and singleton-normalized.

Theorem 13 (closed forms)**.** *For all $p \geq 1$, $k \geq 2$,*

$$\begin{aligned}
\mathrm{Lmin}\big(G(p,k)\big) &= \Big(\tfrac{p^2(k-1)}{p+1},\ \big[\tfrac{p}{p+1}\big]^{p+k}\Big),\\
\mathrm{Lmax}\big(G(p,k)\big) &= \Big(\tfrac{p(k-1)}{k},\ \big[\tfrac{1}{k}\big]^{p},\ \big[\tfrac{p(k-1)}{k}\big]^{k}\Big),\\
\big\|\mathrm{Lmin} - \mathrm{Lmax}\big\|_1 &= \frac{2p\big(p(k-1)-1\big)}{p+1}, \qquad \rho\big(G(p,k)\big) = \frac{2\big(p(k-1)-1\big)}{k(p+1)},
\end{aligned}$$

where the exponents indicate the number of coordinates, listed in the order head, ones, B.

The proof occupies Sections 4.1–4.6: a key cap, the two optimizations, a membership check, and the arithmetic of the gap.

4.1 The key cap

For each $b \in B$,

$$f\big(O \cup \{b\}\big) = \max\big(g(O), h(\{b\})\big) = \max(p,p) = p,$$

so by Theorem 6 every core allocation x satisfies

$$x_1 + \cdots + x_p + x_b \ \leq\ p \qquad \text{for every } b \in B. \tag{KC}$$

That single family of $(p+1)$-player caps drives everything below. Note that the two generators *tie* at these coalitions: the maximum defining f is attained on both sides.

4.2 Leximin

Let $x \in \mathrm{Core}(G(p,k))$ and $m := \min_i x_i$. Applying (KC) with any fixed b gives $(p+1)m \leq p$, so

$$m \leq \frac{p}{p+1}, \qquad \text{hence} \qquad \mu \leq \frac{p}{p+1}.$$

Suppose $m = p/(p+1)$. Then in (KC) the $p+1$ numbers $x_1,\dots,x_p,x_b$ are each at least $p/(p+1)$ and sum to at most $p = (p+1)\cdot p/(p+1)$, so all of them equal $p/(p+1)$. This holds for *every* $b \in B$, hence

$$x_1 = \cdots = x_p = x_{p+1} = \cdots = x_{p+k} = \frac{p}{p+1},$$

and efficiency forces the head to

$$x_0 = T - (p+k)\frac{p}{p+1} = pk - \frac{p(p+k)}{p+1} = \frac{p^2(k-1)}{p+1},$$

using $k-(p+k)/(p+1) = (kp+k-p-k)/(p+1) = p(k-1)/(p+1)$. Thus the max-min allocation is *unique*, and a unique max-min allocation is a fortiori the leximin allocation. Since $p(k-1) \geq 1$, the head's coordinate is at least $p/(p+1)$, so the floor of this allocation is indeed $p/(p+1)$. It remains to check that this point is in the core, which is done in Section 4.4.

4.3 Leximax-min

Let $x \in \mathrm{Core}(G(p,k))$, $M := \max_i x_i$ and $s := x_1 + \cdots + x_p$. By (KC), $x_b \leq p - s$ for every $b \in B$, and trivially $x_b \leq M$ and $x_0 \leq M$. Efficiency gives

$$pk = x_0 + s + \sum_{b \in B} x_b \;\leq\; M + s + k\min(M, p-s). \tag{4}$$

Case A: $M \leq p - s$. Then (4) reads $pk \leq (k+1)M + s$. The case hypothesis is $s \leq p - M$, so $pk \leq (k+1)M + p - M = kM + p$, i.e. $M \geq p(k-1)/k$.

Case B: $M > p - s$. Then (4) reads $pk \leq M + s + k(p-s)$, i.e. $(k-1)s \leq M$, so $s \leq M/(k-1)$. The case hypothesis then gives $M > p - s \geq p - M/(k-1)$, i.e. $Mk/(k-1) > p$, i.e. $M > p(k-1)/k$.

In both cases $M \geq p(k-1)/k$, so the leximax-min ceiling is at least $p(k-1)/k$. Equality can occur only in Case A with every inequality used tight, that is with $x_0 = M$, with $x_b = M$ for all $b \in B$, and with

$$s = pk - (k+1)M = pk - (k+1)\frac{p(k-1)}{k} = \frac{p\big(k^2 - (k+1)(k-1)\big)}{k} = \frac{p}{k}.$$

(Consistency: $p - s = p - p/k = p(k-1)/k = M$, so Case A holds with equality, as it must.) Thus at the min-max level the head and all of B are pinned at $p(k-1)/k$ and the ones share exactly p/k; since $p/k \leq p(k-1)/k$ for $k \geq 2$, the ones occupy the last p positions of the decreasing rearrangement. The next leximax stage minimizes their maximum subject to their sum being p/k, whose unique optimum is $1/k$ each. Hence Lmax is as claimed, again modulo membership.

4.4 Membership

This step is load-bearing: the two optimizations above determine the optima only conditionally on the described allocations being feasible.

Index as in Definition 12 and, for $S \subseteq N$, put

$$\varepsilon := [\,0 \in S\,], \qquad i := |S \cap O|, \qquad j := |S \cap B|,$$

so that $0 \leq i \leq p$, $0 \leq j \leq k$ and

$$g(S) = \varepsilon\, p(k-1) + i, \qquad h(S) = jp, \qquad f(S) = \max\big(\varepsilon\, p(k-1) + i,\; jp\big). \tag{5}$$

Lemma 14. *Both allocations of Theorem 13 lie in* $\mathrm{Core}(G(p,k))$ *for all* $p \geq 1$, $k \geq 2$. *Moreover, for both of them the tight coalitions are exactly*

(T1)	$S = O \cup \{b\},\ b \in B,$	$f(S) = \max(p,p) = p,$
(T2)	$S = \{0\} \cup B',\ \lvert B'\rvert = k-1,$	$f(S) = \max\big(p(k-1), (k-1)p\big) = p(k-1),$
(T3)	$S = N,$	$f(N) = T.$

Proof. The leximin allocation. Write $q := p/(p+1)$ and $H := p^2(k-1)/(p+1) = p(k-1)q$, so $x^* = (H, q, \dots, q)$ with $p+k$ copies of q. Efficiency:

$$H + (p+k)q = q\big(p(k-1) + p + k\big) = q(pk+k) = k(p+1)\frac{p}{p+1} = pk = T,$$

and $x^* > 0$. It remains to check $x^*(S) = \varepsilon H + (i+j)q \le f(S)$.

$\varepsilon = 0$, $j = 0$: S is a subset of O, and $x^*(S) = iq < i = f(S)$ for $i \ge 1$ because $q < 1$; for $i = 0$, $S = \emptyset$. Strict.

$\varepsilon = 0$, $j \ge 1$: bound $f(S) \ge jp$ and use $i \le p$:

$$x^*(S) = (i+j)\frac{p}{p+1} \le (p+j)\frac{p}{p+1} \le jp \iff p + j \le j(p+1) \iff p \le jp,$$

which holds since $j \ge 1$. Equality throughout requires $i = p$ and $j = 1$, i.e. $S = O \cup \{b\}$, which is (T1).

$\varepsilon = 1$, $j \le k-1$: put $G_0 := p(k-1) + i = g(S)$, so $x^*(S) = q(G_0 + j)$ and

$$x^*(S) \le G_0 \iff p(G_0 + j) \le (p+1)G_0 \iff jp \le G_0 = p(k-1) + i,$$

which holds because $jp \le (k-1)p$. Hence $x^*(S) \le g(S) \le f(S)$, with equality exactly when $j = k-1$ and $i = 0$, i.e. $S = \{0\} \cup B'$ with $|B'| = k-1$, which is (T2).

$\varepsilon = 1$, $j = k$: then $h(S) = kp = T$, so $f(S) \ge T \ge x^*(S)$, with equality only for $S = N$ since $x^* > 0$; this is (T3).

The leximax-min allocation. Write $L := p(k-1)/k$ and $r := 1/k$, so $y^* = (L, r, \dots, r, L, \dots, L)$ with p copies of r and k copies of L. Efficiency:

$$L + pr + kL = (k+1)L + \frac{p}{k} = \frac{p\big[(k+1)(k-1)+1\big]}{k} = \frac{pk^2}{k} = pk = T,$$

and $y^* > 0$. Now $y^*(S) = \varepsilon L + ir + jL$.

$\varepsilon = 0$, $j = 0$: $y^*(S) = i/k \le i = f(S)$ since $k \ge 2$, strictly for $i \ge 1$.

$\varepsilon = 0$, $j \ge 1$: comparing with the h-side,

$$\frac{i + jp(k-1)}{k} \le jp \iff i + jp(k-1) \le jpk \iff i \le jp,$$

which holds since $i \le p \le jp$, with equality exactly when $i = p$ and $j = 1$, i.e. (T1).

$\varepsilon = 1$, $j \le k-1$: $y^*(S) = (j+1)L + i/k$, and multiplying by $k > 0$,

$$y^*(S) \le p(k-1) + i \iff (j+1)p(k-1) + i \le kp(k-1) + ki \iff p(j+1-k) \le i,$$

after dividing by $k - 1 \ge 1$; the left-hand side is at most $0 \le i$, with equality exactly when $j = k-1$ and $i = 0$, i.e. (T2).

$\varepsilon = 1$, $j = k$: as before $f(S) \ge h(S) = kp = T \ge y^*(S)$, with equality only at $S = N$. □

4.5 The hinge: both selections saturate the same caps

Lemma 14 contains the expository point of the paper. The active constraints are the same for Lmin and Lmax: the two families (T1) and (T2), plus efficiency. Both are coalitions at which the two generators *tie*, which is the structural reason they are the active ones — there is no slack on either

side to give. What differs is only how the two selections distribute mass *inside* a tight coalition. On $S = O \cup \{b\}$, which has total exactly p:

	each of the p ones	the single $b \in B$
Lmin :	$\dfrac{p}{p+1}$	$\dfrac{p}{p+1}$
Lmax :	$\dfrac{1}{k}$	$\dfrac{p(k-1)}{k}$

Leximin splits the budget p evenly over all $p+1$ members; leximax-min gives the ones the least it can and puts the whole remainder on the one member of B, because loading B is how it lowers the global maximum. The divergence is visible in this one coalition. Everything else is bookkeeping: leximin's even split leaves the residual $T - (p+k)p/(p+1)$ with nowhere to go but the head, while leximax-min's loading of every $b \in B$ up to $p(k-1)/k$ consumes almost all of T across B.

4.6 The gap

Proof of Theorem 13, conclusion. Put $D := p(k-1) - 1 \geq 0$. Coordinatewise,

$$\begin{aligned}
\text{head:} \qquad & \frac{p^2(k-1)}{p+1} - \frac{p(k-1)}{k} = \frac{p(k-1)D}{k(p+1)}, \\
\text{each of the } p \text{ ones:} \qquad & \frac{p}{p+1} - \frac{1}{k} = \frac{D}{k(p+1)}, \\
\text{each of the } k \text{ members of } B: \qquad & \frac{p(k-1)}{k} - \frac{p}{p+1} = \frac{pD}{k(p+1)},
\end{aligned}$$

all three nonnegative. Hence

$$\| \operatorname{Lmin} - \operatorname{Lmax} \|_1 = \frac{p(k-1)D + pD + kpD}{k(p+1)} = \frac{D\big(p(k-1) + p + kp\big)}{k(p+1)} = \frac{2pkD}{k(p+1)} = \frac{2pD}{p+1},$$

and $\rho = 2pD/\big((p+1)pk\big) = 2D/\big(k(p+1)\big)$. □

Corollary 15 (deficit identity)**.** $2 - \rho\big(G(p,k)\big) = \dfrac{2n}{k(p+1)}$, *where* $n = p+1+k$.

Proof. $2 - \dfrac{2\big(p(k-1)-1\big)}{k(p+1)} = \dfrac{2\big[k(p+1) - p(k-1) + 1\big]}{k(p+1)} = \dfrac{2(kp + k - pk + p + 1)}{k(p+1)} = \dfrac{2(k+p+1)}{k(p+1)}.$ □

Remark 16 (the deficit is twice the overlap)**.** *Corollary 15 also follows from Lemma 3. In $G(p,k)$ the coordinatewise minima of the two allocations are $p(k-1)/k$ at the head, $1/k$ at each one, and $p/(p+1)$ at each member of B, so*

$$\sum_i \min\big(\operatorname{Lmin}_i, \operatorname{Lmax}_i\big) = \frac{p(k-1)}{k} + \frac{p}{k} + \frac{kp}{p+1} = p + \frac{kp}{p+1} < p + k = n - 1,$$

and $2 \cdot \big(p + kp/(p+1)\big)/(pk) = 2(p+1+k)/\big(k(p+1)\big)$. The overlap between the two selections is thus bounded by n while $T = pk$ grows quadratically: this is the precise sense in which their supports become asymptotically disjoint.

n	(p,k)	lower bound on $s(n)$	decimal
6	$(2,3)$	$2/3$	0.6667
7	$(2,4)$	$5/6$	0.8333
8	$(3,4)$	1	1.0000
9	$(3,5)$	$11/10$	1.1000
10	$(4,5)$	$6/5$	1.2000
11	$(4,6)$	$19/15$	1.2667
12	$(5,6)$	$4/3$	1.3333
16	$(7,8)$	$3/2$	1.5000
20	$(9,10)$	$8/5$	1.6000
100	$(49,50)$	$48/25$	1.9200

Table 1: The bound of Theorem 17 at the balanced split. All entries are exact; decimals are rounded.

4.7 The supremum

Theorem 17 (value of the supremum)**.** *For every $n \geq 4$,*

$$s(n) \;\geq\; 2 - \frac{8n}{n^2 - (n \bmod 2)} = \begin{cases} 2 - \dfrac{8}{n}, & n \text{ even}, \\ 2 - \dfrac{8n}{n^2-1}, & n \text{ odd}, \end{cases}$$

attained on the two-generator class by $G(p,k)$ with the balanced split $k = \lceil n/2 \rceil$, $p+1 = \lfloor n/2 \rfloor$. Consequently $\sup_n s(n) = 2$.

Proof. By Corollary 15 the deficit of $G(p,k)$ is $2n/(k(p+1))$ with $k+(p+1) = n$, $k \geq 2$ and $p+1 \geq 2$. Minimizing the deficit means maximizing the product $k(p+1)$ of two positive integers with fixed sum n, whose maximum is $\lfloor n/2 \rfloor \lceil n/2 \rceil$, equal to $n^2/4$ for even n and $(n^2-1)/4$ for odd n, attained at the balanced split (for $n \geq 4$ this split satisfies the constraints $k \geq 2$, $p \geq 1$). Substituting gives the displayed bound. For the last claim, $\rho \leq 2$ always by Lemma 3, and $2 - 8/n \to 2$. □

Table 1 lists the resulting values. The ratio reaches 1 at $n = 8$ and passes it at $n = 9$: from eight players on the two egalitarian core selections can differ by the entire worth of the grand coalition, and from nine players by more.

Remark 18 (the degenerate corner)**.** *$G(1,2)$ has $n = 4$, $T = 2$, $g = (1,1,0,0)$, $h = (0,0,1,1)$ and $\rho = 2(1 \cdot 1 - 1)/(2 \cdot 2) = 0$, consistent with the four-player rigidity; the deficit identity reads $2 - 0 = 2n/(k(p+1)) = 8/4 = 2$.*

Remark 19 (there is nothing special about $|B| = 4$)**.** *An earlier version of this family fixed $k = |B| = 4$, giving $\rho = (3p-1)/\big(2(p+1)\big) \to 3/2$. Corollary 15 shows that k is a free parameter and that the optimum is $k \approx n/2$, so the limit $3/2$ was an artifact of the fixed block size; the $k = 4$ slice is already beaten from $n = 10$ on ($7/6$ against $6/5$). The number 4 that appears as $|\operatorname{supp}(\mathrm{Lmax})|$ in the $n = 5$ extremal class (Section 5) is a small-n phenomenon and not a feature of the asymptotics: in $G(p,k)$ every coordinate of* Lmax *is strictly positive, so $|\operatorname{supp}(\mathrm{Lmax})| = n$ and no player is starved at all.*

n	totals	games	enumerated max ρ	Theorem 17
3	2–30	225	0	—
4	2–30	3,040	0	0
5	2–24	4,357	1/2	1/3
6	2–20	8,213	2/3	2/3
7	2–14	1,609	5/6	5/6
8	2–12	1,049	1	1

Table 2: Complete two-generator enumerations. The rows $n = 3, 4$ are controls: they reproduce the four-player rigidity. The last column is the value of the balanced-split family bound.

5 Fixed n: enumeration, certificates, and the $n = 5$ anomaly

5.1 Complete two-generator enumeration

Write $s_2(n)$ for the supremum of ρ over singleton-normalized two-generator exact games on n players. Table 2 reports complete enumerations, up to player permutation, generator swap and scaling, of all integer generator pairs with common total in the stated range. All arithmetic is exact rational.

Two readings must be kept apart.

What is complete. Because ρ is scale-invariant, every two-generator game with rational generators has a unique primitive integer representative, and enumeration to total $T_{\max}$ is exactly complete for the games whose primitive representative has total at most $T_{\max}$ — equivalently, for rational shapes of denominator at most $T_{\max}$. It says nothing about finer denominators or about irrational generators, so enumeration-to-T is a lower-bound method for $s_2(n)$: the entries of Table 2 are maxima over the enumerated range, not values of s_2. As a partial probe of the excluded region, shape hill-climbing at fine resolution was run from the known witnesses (resolution 1/60 at $n = 5$; 1/56 and 1/60 at $n = 6$; 1/64 at $n = 7$). None improved on the tabulated maximum; at $n = 7$ the climb from the 5/6 witness stalled after 42 steps, and a second start from a base of 2/3 climbed only to 3/4.

What agrees. The enumerated maxima agree exactly with Theorem 17 at $n = 6, 7, 8$, so on the two-generator class that bound is attained there, and

$$s_2(n) \stackrel{?}{=} 2 - \frac{8n}{n^2 - (n \bmod 2)} \qquad (n \geq 6)$$

is a live candidate, currently confirmed at three values of n. At $n = 5$ the two disagree, and the enumeration is the correct one; see Section 5.3.

5.2 Searching the full exact class

The two-generator restriction is not innocuous. Adding a generator *lowers* the envelope and hence *grows* the core, so it is not automatic that two generators are extremal. The following observation lets one search the entire exact class rather than a bounded-generator slice.

Lemma 20 (core closing)**.** *Fix $T > 0$ and a vector $c = (c_S)$ indexed by the proper coalitions, and put $P(c) = \{x : x(N) = T,\ x(S) \geq c_S \text{ for all proper } S\}$. If $P(c) \neq \emptyset$, then the game v defined by $v(S) = \min_{x \in P(c)} x(S)$ for proper S and $v(N) = T$ is exact, and $\mathrm{Core}(v) = P(c)$.*

Proof. Every $x \in P(c)$ satisfies $x(S) \geq v(S)$ by definition of v and $x(N) = T$, so $P(c) \subseteq \mathrm{Core}(v)$; and each minimum is attained, so v is exact. Conversely, $v(S) \geq c_S$ for every proper S because every point of $P(c)$ satisfies $x(S) \geq c_S$; hence $x \in \mathrm{Core}(v)$ implies $x(S) \geq v(S) \geq c_S$ and $x(N) = T$, i.e. $x \in P(c)$. □

Perturbing c and re-closing therefore moves within the exact class and reaches cores with arbitrarily many vertices. Hill climbing in this "c-space", *seeded at the extremal two-generator game* of Table 2, produced no improvement at $n = 5$ (240 steps, stays at $1/2$), $n = 6$ (496 steps, stays at $2/3$) or $n = 7$ (301 steps, stays at $5/6$). Since the search is over all exact games, this certifies the two-generator optimum as a *local* optimum in the full exact class at those three values of n. Independently, from the known extremal at each of $n = 5, 6, 7, 8$, about 200 structured and random third generators at fixed common total were added: zero improvements.

One limitation must be stated. Unseeded c-space restarts reach only $1/3$ at $n = 6$, far below the known $2/3$; that variant of the search is under-powered for $n \geq 6$ and certifies nothing. Only the seeded runs support the local-optimality claim, and none of these runs bears on global optimality.

5.3 The $n = 5$ anomaly

At $n = 5$ the balanced-split family gives only $1/3$, but the truth is $1/2$, and the witness has a non-uniform h-block. The proof is by hand from two flat splits of the game; no linear program and no vertex enumeration is used.

Theorem 21 ($n = 5$)**.** *Let w be the lower envelope of $g^1 = (1, 0, 0, 1, 2)$ and $g^2 = (0, 1, 3, 0, 0)$. Then w is exact and singleton-normalized with $w(N) = 4$, and*

$$\mathrm{Lmin}(w) = \left(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{5}{4}, \tfrac{1}{2}, \tfrac{5}{4}\right), \qquad \mathrm{Lmax}(w) = (1, 0, 1, 1, 1), \qquad \| \mathrm{Lmin} - \mathrm{Lmax} \|_1 = 2, \qquad \rho(w) = \tfrac{1}{2}.$$

In particular $s(5) \geq 1/2 > 1/3$.

Proof. Exactness and normalization are immediate from Lemma 1, each coordinate vanishing in one generator. Two flat splits control the core: $w(\{1, 2\}) = 1$ and $w(\{3, 4, 5\}) = 3$ sum to $w(N) = 4$, so $x_1 + x_2 = 1$ on all of $\mathrm{Core}(w)$, and likewise $w(\{2, 4\}) = 1$ with $w(\{1, 3, 5\}) = 3$ gives $x_2 + x_4 = 1$. Putting $s := x_1$, every core point therefore satisfies

$$x_1 = x_4 = s, \qquad x_2 = 1 - s, \qquad x_3 + x_5 = 3 - s, \tag{6}$$

with $s \in [0, 1]$ by nonnegativity of x_1 and x_2.

Leximin. By (6), $\min_i x_i \leq \min(s, 1 - s) \leq 1/2$, with equality forcing $s = 1/2$ and hence $x_1 = x_2 = x_4 = 1/2$. Then $x_3 + x_5 = 5/2$, and the leximin continuation splits this evenly, giving $5/4$ each. The resulting allocation $x = (1/2, 1/2, 5/4, 1/2, 5/4)$ lies in the core, as direct substitution against both generators confirms. Hence $\mathrm{Lmin}(w)$ is as claimed.

Leximax-min. By (6), $\max_i x_i \geq (x_3 + x_5)/2 = (3 - s)/2 \geq 1$ since $s \leq 1$. Equality forces $s = 1$, so $x = (1, 0, x_3, 1, x_5)$ with $x_3 + x_5 = 2$; the ceiling $\max_i x_i = 1$ then forces $x_3 = x_5 = 1$. The unique candidate $(1, 0, 1, 1, 1)$ lies in the core, so $\mathrm{Lmax}(w) = (1, 0, 1, 1, 1)$.

Finally $\| \mathrm{Lmin} - \mathrm{Lmax} \|_1 = \frac{1}{2} + \frac{1}{2} + \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 2$ and $\rho = 2/4 = 1/2$. □

Four independent methods agree on this game: the hand argument above, exact rational core-vertex enumeration, a staged exact solver over linear real arithmetic, and a from-scratch exact-rational two-phase simplex with Bland's rule.

The value $\frac{1}{2}$ is in fact the supremum: $s(5) = \frac{1}{2}$ (Theorem 33 below, by a configuration-complete exact finite check). Before that machinery, the searches already pointed there: roughly 4×10^4 distinct exact five-player games evaluated in exact rational arithmetic produced nothing above $1/2$ — the complete two-generator enumeration of Table 2, complete three-generator enumerations at totals 4–7, c-space hill climbing in the sense of Lemma 20, and fine-denominator shape climbing from the extremal witness — with the $n = 3, 4$ controls returning maximum 0 from the same pipeline.

Structure of the extremal class. The following are computational findings over 360 distinct profiled games at $n = 5$, of which 199 have $\rho > 1/3$ and 107 attain $\rho = 1/2$. Among the 107 extremal games, without exception:

- Lmin has stratum composition $(3, 2)$, and Lmax has stratum composition $(1, 4)$;
- Lmax has exactly one zero coordinate, so $|\operatorname{supp}(\text{Lmax})| = 4$;
- normalized to $T = 4$ the pair is unique up to permutation: Lmin $\sim (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{5}{4}, \frac{5}{4})$ and Lmax $\sim (0, 1, 1, 1, 1)$ — that is, exactly the pair of Theorem 21.

Among the 199 games above $1/3$, the Lmin strata are $(3, 2)$ in 196 cases and $(2, 1, 2)$ in 3; the pattern $(4, 1)$ never occurs above $1/3$, and Lmax is never the uniform allocation.

One tempting invariant is refuted. Call S a *flat split* if $v(S) + v(N \setminus S) = v(N)$; then $x(S)$ is constant on the core, which is the affine seesaw visible in (6) — in the game of Theorem 21, $w(\{1, 2\}) + w(\{3, 4, 5\}) = 1 + 3 = 4$. Only 76 of the 107 extremal games have any flat split, and 31 attain exactly $1/2$ with none, so the seesaw is a feature of the particular presentation in Theorem 21 and not an invariant of extremality. When a flat split does exist its sizes are always $\{2, 3\}$, and the Lmax-starved player lies on the size-2 side in all 76 cases.

Two regimes. Theorem 17 gives only $1/3$ at $n = 5$ because the family $G(p, k)$ has a uniform h-block and the available splits $(k, p + 1) \in \{(2, 3), (3, 2)\}$ are too coarse; the extremal five-player game has a non-uniform h-block. The crossover to the G-family shape happens at $n = 6$. The two regimes differ qualitatively and not just numerically: at $n = 5$ the extremal Lmax starves one player to 0 and has support of size 4, whereas in $G(p, k)$ the extremal Lmax is strictly positive everywhere. Explaining the crossover is open (Section 7).

5.4 One-coalition certificates and the regime change

Write, for any singleton-normalized exact game with the general dual bound $f(S) = T - v(N \setminus S)$,

$$\lambda := \min_{\emptyset \neq S} \frac{f(S)}{|S|}, \qquad \nu := \max_{\emptyset \neq S} \frac{v(S)}{|S|}, \qquad \mu := \min_i \text{Lmin}(v)_i, \qquad M := \max_i \text{Lmax}(v)_i.$$

The general duality gives the bracketing $\mu \leq \lambda$ and $M \geq \nu$ at once (for $\mu \leq \lambda$ the proof of Theorem 7 applies verbatim). On a two-generator game the right-hand side of (3) is exactly ν, since $v(P \cup Q) = \min(g(P), h(Q))$. Observation 10 says the second bracket can be strict from $n = 6$ on. Two facts sharpen this into a statement about the regime change at five players.

Proposition 22 (covering obstruction)**.** *Let $\tau_\nu := \{S : v(S) = \nu|S|\}$ be the ν-tight family. If τ_ν covers N and $\nu > T/n$, then no core point satisfies $x \leq \nu \mathbf{1}$, and hence $M > \nu$.*

Proof. Sum the core inequalities $x(S) \geq v(S) = \nu|S|$ over a cover with weights $w \geq 0$: with $c_i := \sum_{S \ni i} w_S \geq 1$ this gives $\sum_i c_i(x_i - \nu) \geq 0$. If $x \leq \nu \mathbf{1}$, every term is nonpositive, so $x_i = \nu$ for every i; then $T = \nu n$, contradicting $\nu > T/n$. □

Moreover no *balanced* collection can witness $M > \nu$: for balanced w, $\sum_S w_S v(S) \le T$ because the core is nonempty, and the same computation then yields $\sum_S w_S\big(\nu|S| - v(S)\big) \ge \nu n - T$, which is what equality requires. So a failure of $M = \nu$ needs a strictly over-covering fractional cover: no balanced collection can witness one. A covering ν-tight family is one way this arises, and it is what occurs at the smallest counterexample of Observation 10: the four ν-tight sets $\{2,3\},\{1,3,4\},\{1,3,5\},\{1,3,6\}$ cover N with per-player excess $(2,0,3,0,0,0)$, and player 2, lying only in $\{2,3\}$, blocks every balanced subfamily. At the five-player extremal game of Theorem 21, by contrast, $\nu = 1$ and the ν-tight family $\{3,5\},\{1,3,5\},\{3,4,5\}$ misses player 2, so Proposition 22 does not apply, and the dual minimum $h(N) = 1$ is attained at the balanced partition $\{2,4\} \mid \{1,3,5\}$. The covering test is sufficient for $M > \nu$ but not necessary: at six players, games with $M > \nu$ outnumber those with a covering ν-tight family.

Theorem 23 (one-coalition certificates at five players)**.** *For every singleton-normalized exact game on five players, $\mu = \lambda$ and $M = \nu$: both egalitarian levels are certified by a single coalition.*

Proof, by finite exact check. By the duality above, $M = \nu$ fails if and only if some fractional cover w has $\sum_S w_S h(S) < h(N)$, where $h(S) = \nu|S| - v(S) \ge 0$; likewise for μ with $f'(S) = f(S) - \lambda|S|$. Both objectives are nonnegative on the cover polyhedron, which is pointed with recession cone the nonnegative orthant, so each minimum is attained at a vertex; on five players there are 2287 vertices, each of support at most five. Fixing a vertex and a coalition attaining ν (resp. λ) makes every hypothesis, and the failure inequality, linear in v, and $v(N) = 1$ is free by homogeneity. Exactness is imposed through the 280 inequalities that the facet theorem of Studený and Kratochvíl (2022) assigns to the sixteen indecomposable min-semi-balanced systems on five players under all permutations; this description was validated in both directions against independent exactness tests. Each of the 1142 symmetry classes of (vertex, anchor) pairs, on each side, is infeasible in exact rational arithmetic. □

Two remarks calibrate the proof. First, deleting the 280 exact-cone inequalities makes 361 (resp. 339) of the over-covering classes feasible, so it is exactness — not the shape of the certificate — that forces the one-coalition property at five players. Second, the M-side is false from six players on (Observation 10), and the same reduction detects that failure. The theorem is therefore the formal content of the regime change of Section 5.3: at five players both selections are governed by single bottleneck coalitions, while from six players the ceiling requires genuinely fractional covers — which is precisely where $s(n)$ begins its climb toward 2. One consequence localizes all the difficulty of $s(5)$.

Corollary 24 (floor from ceiling)**.** *For every singleton-normalized exact five-player game, $\mu \ge \min_i \mathrm{Lmax}(v)_i$.*

Proof. Lmax is a core point, so $f(S) = T - v(N \setminus S) \ge T - \mathrm{Lmax}(N \setminus S) = \mathrm{Lmax}(S)$ for every S; hence $\lambda \ge \min_S \mathrm{Lmax}(S)/|S| \ge \min_i \mathrm{Lmax}_i$, and $\mu = \lambda$ by Theorem 23. □

If Lmax has no zero coordinate, the leximin floor is therefore bounded below for free and elementary gap arithmetic bounds the ratio; every genuinely hard configuration for $s(5)$ has a *zero* in Lmax. The theorem does *not* by itself deliver the starvation bound needed for those configurations; that bound is a statement about a different functional and requires its own argument, which the next theorem supplies in the extremal one.

Theorem 25 (the extremal configuration is pinned)**.** *Let v be a singleton-normalized exact five-player game whose* Lmin *strata form the pattern $(3,2)$ and whose* Lmax *strata form the pattern $(1,4)$ with the singleton stratum at zero, at player j. Then*

$$f(\{j\}) = T - v(N \setminus \{j\}) \;\ge\; \tfrac{T}{8}.$$

Proof. Normalize $T = 1$ and suppose $f(\{j\}) < \frac{1}{8}$. The starved player j is Lmin-poor by Corollary 30 below (whose two-line proof uses only Lemma 29 and is independent of this theorem); write $P = \{j, p_1, p_2\}$ for the poor stratum and $R = \{r_1, r_2\}$ for the rich pair, with $x = \text{Lmin}$ equal to μ on P and $\varrho = (1 - 3\mu)/2$ on R, and $z = \text{Lmax}$ equal to 0 at j and $\frac{1}{4}$ elsewhere. Call S *f-tight* when $x(S) = f(S)$, equivalently when $N \setminus S$ is v-tight at x.

Step 0. $\lambda \le f(\{j\}) < \frac{1}{8}$, and $\mu \le \lambda$ holds for every game (the easy half of Theorem 23, i.e. Theorem 7's argument), so $\mu < \frac{1}{8}$; hence $\varrho > \frac{5}{16} > \frac{1}{4}$ and $c := 1 - 5\mu > 0$.

Step 1 (key lemma). Every f-tight S satisfies $|S \cap \{p_1, p_2\}| \le |S \cap R|$. Indeed, $U = N \setminus S$ is v-tight, and $z \in \text{Core}(v)$ gives, with $a = |U \cap P|$, $b = |U \cap R|$, $\varepsilon = [\, j \in U \,]$,

$$\mu a + \varrho b \;=\; x(U) = v(U) \;\le\; z(U) \;=\; \tfrac{a+b-\varepsilon}{4}.$$

The claim is $b \le a - \varepsilon$. A violation leaves four shapes, each contradictory: $a - \varepsilon = 0$ with $\varepsilon = 0$ gives $\varrho \le \frac{1}{4}$; with $\varepsilon = 1$ it gives $\mu \le b(\frac{1}{4} - \varrho) < 0$; $a - \varepsilon = 1$ with $\varepsilon = 1$ (so $b = 2$) gives $1 - \mu \le \frac{3}{4}$, i.e. $\mu \ge \frac{1}{4}$; and $a - \varepsilon = 1$ with $\varepsilon = 0$, $b = 2$ — the coalition $U = \{p_i\} \cup R$ — gives $1 - 2\mu \le \frac{3}{4}$, i.e. $\mu \ge \frac{1}{8}$, against the failure hypothesis. This last case is the only one that mentions $\frac{1}{8}$.

Step 2 (the pinning, dualized). That P is the Lmin bottom stratum means every core point $y \ge \mu\mathbf{1}$ has $y(P) = 3\mu$. Substituting $u = y - \mu\mathbf{1}$, this is $\max\{u(P) : u \ge 0,\ u(N) = c,\ u(S) \le h(S)\} = 0$ with $h(S) := f(S) - \mu|S| \ge 0$. LP duality supplies α and $\beta \ge 0$ with $\alpha + B_k \ge [\, k \in P \,]$ for every player k, where $B_k = \sum_{S \ni k} \beta_S$, and $\alpha c + \sum_S \beta_S h(S) \le 0$. Since $c > 0$ and $h \ge 0$, necessarily $\alpha \le 0$; put $A = -\alpha \ge 0$, so $B_k \ge 1 + A$ on P and $B_k \ge A$ on R.

Step 3 (exact tightness). $x \in \text{Core}(v)$ gives $h(S) \ge x(S) - \mu|S| = (\varrho - \mu)\,|S \cap R|$, so

$$\sum_S \beta_S h(S) \;\ge\; (\varrho - \mu)(B_{r_1} + B_{r_2}) \;\ge\; 2A(\varrho - \mu) \;=\; cA,$$

using $2(\varrho - \mu) = 1 - 5\mu = c$ *exactly*. Step 2 bounds the same sum by cA, so both inequalities are equalities: $B_{r_1} + B_{r_2} = 2A$, and every support set of β is f-tight.

Step 4 (counting). The key lemma now applies to every support set, so $B_{p_1} + B_{p_2} = \sum_S \beta_S |S \cap \{p_1, p_2\}| \le \sum_S \beta_S |S \cap R| = 2A$, while Step 2 demands $B_{p_1} + B_{p_2} \ge 2 + 2A$. Contradiction. □

Remark 26 (sharpness, and the archived check)**.** *The constant is not inserted; it falls out. Exactly one line of the proof mentions* $\frac{1}{8}$ *— the coalition shape* $U = \{p_i\} \cup R$ *— and at* $\mu = T/8$ *that inequality becomes an equality and the key lemma fails at* $S = \{j, p_{i'}\}$*. This is realized: in the extremal game of Theorem 21 (which lies in the configuration with* $\mu = T/8$*) the* f*-tight sets* $\{1, 2\}$ *and* $\{2, 4\}$ *are precisely of that shape. The proof also consumes strictly less than the original finite check did: neither the hard direction of Theorem 23 nor the facet description appears — only core membership of the two selections, the floor bracketing* $\mu \le \lambda$*, Corollary 30, and the pinning. The original check (*10,974 *exact programs over the* 354 *pinning-dual vertices) remains archived as an independent confirmation.*

Corollary 27 (the pinned solution pair)**.** *In the configuration of Theorem 25, suppose in addition that the starved player is* Lmin*-poor and that some coalition attaining* ν *has three players. Then, up to relabelling,* $\text{Lmin} = \left(\frac{T}{8}, \frac{T}{8}, \frac{T}{8}, \frac{5T}{16}, \frac{5T}{16}\right)$*,* $\text{Lmax} = \left(0, \frac{T}{4}, \frac{T}{4}, \frac{T}{4}, \frac{T}{4}\right)$*, and* $\rho(v) = \frac{1}{2}$ *exactly.*

Proof. Normalize $T = 1$. The Lmax pattern gives $M = 1/4$ directly. Core membership of the Lmax vector gives, for $S \ne \{j\}$, $f(S) \ge \big(|S| - [\, j \in S \,]\big)/4 \ge |S|/8$, and Theorem 25 gives $f(\{j\}) \ge 1/8$, so $\lambda \ge 1/8$. By the M-side of Theorem 23, $\nu = M = 1/4$, so a three-player coalition S_0 attaining ν has $v(S_0) = 3/4$ and $f(N \setminus S_0) = 1/4$ on two players, so $\lambda \le 1/8$; hence $\lambda = 1/8$, and Theorem 23 gives $\mu = 1/8$. With the starved player Lmin-poor, $\mu = 1/8 \le 1/6$ gives $(1 - 3\mu)/2 \ge 1/4 = M$, so the coordinatewise gap telescopes to $\|\,\text{Lmin} - \text{Lmax}\,\|_1 = T - 4\mu = T/2$, and the two displayed allocations are the unique ones consistent with the strata, the floor and the totals. □

The first additional hypothesis is automatic: Corollary 30 below proves, in two lines, that the starved player is always Lmin-poor. The second — a three-player coalition attaining ν — holds in every configuration game in the combined harvests (53 of 53) but remains open in general, and is recorded in Section 7.

For the ratio *bound*, as opposed to the pinned pair, both hypotheses can in fact be dispensed with.

Theorem 28 (unconditional configuration bound)**.** *In the configuration of Theorem 25,* $\rho(v) \le \frac{1}{2}$.

Proof, by gap arithmetic. Normalize $T = 1$; the Lmax pattern gives $M = 1/4$ and the $(3, 2)$ pattern gives $\mathrm{Lmin} = (\mu, \mu, \mu, \varrho, \varrho)$ with $\varrho = (1 - 3\mu)/2$. The starved player is Lmin-poor by Corollary 30 below (a two-line consequence of Lemma 29, independent of this theorem), so the coordinatewise gap is $1 - 4\mu$ for $\mu \le 1/6$ and 2μ otherwise, and Theorem 25 gives $\mu \ge 1/8$, so both branches are at most $\frac{1}{2}$. (Historically the Lmin-rich branch, whose gap $3/2 - 6\mu$ would exceed $\frac{1}{2}$ for $\mu < 1/6$, was first closed by a finite check of the same design as Theorem 25 — the rich relabelling of the pinning polyhedron, 31×354 exact programs, all infeasible, with the drop-the-pinning control restoring feasibility. The corollary supersedes that check by showing the branch is vacuous; the check remains archived as an independent confirmation.) □

Two calibrations. First, the floor bound is not removable in favour of soft arguments: from Theorem 23 and core membership of both allocation vectors alone — everything that is game-free — the best obtainable bound on ρ at five players is exactly $\frac{3}{2}$, so every configuration needs a lower bound on the floor of the strength of Theorem 25. Second, the attempted refutation of the ν-triple hypothesis produced, once again, a certificate whose audited model is a genuine exact game *outside* the configuration (a $(1, 4)$-shaped Lmax with minimum $1/30$, ratio $5/12$) — the third certificate-not-realizable instance of this program, and the reason hypotheses are retired here only by finite checks, never by feasible certificates alone.

The configuration hypothesis is essential, not decorative: dropping the $(3, 2)$ pinning makes the same programs satisfiable, and their models are genuine exact five-player games with $T - v(N \setminus \{i\}) = T/16$ for some player i — half the bound — but with Lmin = Lmax, hence no separation at all. The cases outside Corollary 27 — the Lmax stratum shapes $(1, 2, 2)$ and $(1, 1, 3)$, the $(4, 1)$ Lmin boundary, and the Lmin-rich starved player — are closed in the next subsection, which settles the supremum in full.

5.5 The five-player supremum in full

Three localization lemmas come first; all are consequences of Theorem 23 together with core membership of the two allocation vectors, and none needs a computation.

Lemma 29 (anchor localization)**.** *Let v be a singleton-normalized exact five-player game, S_0 a coalition attaining ν and S_* a coalition attaining λ (both exist by Theorem 23). Then*

(a) $|S_0| \ge 2$, *and* $\mathrm{Lmax}_i = M$ *for every* $i \in S_0$; *consequently the* Lmax *top stratum contains at least two players;*

(b) $\mathrm{Lmin}_i = \mu$ *for every* $i \in S_*$, *and* $\mathrm{Lmax}(S_*) \le \mu|S_*|$; *consequently* $\mu \ge \min_{i \in P} \mathrm{Lmax}_i$, *where P is the* Lmin *bottom stratum;*

(c) $2M + 3\mu \le T$.

Proof. (a) If $S_0 = \{i\}$ then $M = \nu = v(\{i\}) = 0$, but $\mathrm{Lmax}(N) = T > 0$ forces $M \ge T/5$. For the second claim, $\mathrm{Lmax}(S_0) \ge v(S_0) = M|S_0|$ by core membership while $\mathrm{Lmax}_i \le M$ throughout, so Lmax is constant M on S_0. (b) Core membership gives $\mathrm{Lmin}(S_*) \le f(S_*) = \mu|S_*|$ while $\mathrm{Lmin}_i \ge \mu$

gives the reverse, so $\mathrm{Lmin} = \mu$ on S_*; the same cap applied to Lmax gives $\mathrm{Lmax}(S_*) \le \mu|S_*|$, whence $\mu \ge \mathrm{Lmax}(S_*)/|S_*| \ge \min_{i\in S_*} \mathrm{Lmax}_i \ge \min_{i\in P} \mathrm{Lmax}_i$. (c) By the floor side, $f(N\setminus S_0) \ge \mu(5-|S_0|)$, i.e. $M|S_0| + \mu(5-|S_0|) \le T$; since $M \ge T/5 \ge \mu$ the left side is nondecreasing in $|S_0|$, and $|S_0| = 2$ is admissible by (a). □

Part (b) sharpens Corollary 24: the minimum ranges over the Lmin-*poor* players only. Combined with the trivial $\mu \le T/5$ it closes, by two lines, the hypothesis that resisted the configuration analysis.

Corollary 30 (the starved player is Lmin-poor)**.** *In the configuration of Theorem 25 —* Lmax *strata* $(1, 4)$ *with the singleton at zero, at player* j *— the player* j *lies in the* Lmin *bottom stratum.*

Proof. The stratum pattern forces $4M = T$. If $j \notin P$ then $\mathrm{Lmax}_i = M = T/4$ for every $i \in P$, so Lemma 29(b) gives $\mu \ge T/4$, contradicting $\mu \le T/5$. □

In particular the Lmin-rich branch of Theorem 28's proof is vacuous, the first additional hypothesis of Corollary 27 is automatic, and the corresponding finite check (31×354 programs) is superseded by the two lines above.

The second ingredient is an exact certificate form for the two selections, replacing anchors and pinnings by linear identities.

Lemma 31 (staged Farkas certificates)**.** *Let* v *be a game with nonempty core,* ϕ *a core point with level sets* $B_1 \prec \cdots \prec B_k$ *in increasing order of payoff. Then* $\phi = \mathrm{Lmin}(v)$ *if and only if for every stage* t *and every* $i \in B_t$ *there exist multipliers* $y_A \ge 0$ *supported on proper* ϕ*-tight coalitions,* $\eta \in \mathbb{R}$*,* $\rho_l \in \mathbb{R}$ *for* $l \in B_1 \cup \cdots \cup B_{t-1}$ *and* $\sigma_c \ge 0$ *for* $c \in B_t$ *with, for every player* j*,*

$$-\sum_{A\ni j} y_A + \eta + [\,j \in B_{<t}\,]\rho_j - [\,j \in B_t\,]\sigma_j \;=\; [\,j = i\,].$$

The mirror statement characterizes Lmax*, with the level sets taken in decreasing order,* σ *entering with a plus sign, and right-hand side* $-[\,j = i\,]$*.*

Proof sketch. Leximin is computed stagewise: with the earlier strata pinned, the level μ_t is the largest uniform lower bound on the unpinned players, and B_t collects the players whose maximum subject to that equals μ_t. $\phi = \mathrm{Lmin}(v)$ is equivalent to the family of stage conditions $\max\{x_i : x \in \mathrm{Core}(v),\ x_l = \phi_l\ (l \in B_{<t}),\ x_c \ge \mu_t\ (c \in B_{\ge t})\} = \phi_i$ for $i \in B_t$. Each is a linear program whose feasible set contains ϕ; the constraints active at ϕ are efficiency, the ϕ-tight coalitions, the pinned equalities, and the level bounds on B_t itself. Dualizing over the tangent cone at ϕ and computing the polar cone gives exactly the displayed identity; complementary slackness confines the support of y to tight coalitions and eliminates the bilinear terms $y_A v(A)$, which is what makes the certificate linear in (v, ϕ, y) jointly. The converse direction is the same computation read backwards. The certificate was derived independently twice within the program (once per player and stage as displayed, once per payoff cut) and the two forms verified against each other in both directions on 30 games and 6 non-optimal core points; the mirror was re-derived from scratch and validated on 1050 games (5,250 ceiling certificates, 10,500 counting the floor form, no failures), with negative controls rejecting a false Lmax claim and the core midpoint 261/261 times each. □

With these in place the supremum closes. Recall that a *configuration* is a triple: the Lmin stratum composition, the Lmax stratum composition, and the table of overlaps between the two stratifications. At $n = 5$ there are 2961.

Theorem 32 (the $(4,1)$ regime)**.** *Every singleton-normalized exact five-player game whose* Lmin *strata form the pattern* $(4,1)$ *satisfies* $\rho(v) \le \frac{1}{3}$*, and the bound is attained.*

Theorem 33 (five-player supremum)**.** *Every singleton-normalized exact five-player game satisfies* $\rho(v) \le \frac{1}{2}$*. With Theorem 21,* $s(5) = \frac{1}{2}$*.*

Proof of both, by configuration-complete finite check. Relabelling players reduces every game to the canonical layout of its configuration, so it suffices to exclude $\rho > \frac{1}{2}$ (resp. $\rho > \frac{1}{3}$ on the $(4,1)$ class) in each configuration. This was done by two pipelines sharing no machinery, with machine-audited coverage: every one of the 2961 configurations lands in exactly one bucket, no run errored, and none was left feasible.

The first pipeline is structural: 1953 configurations are impossible by Lemma 29(a) alone (Lmax composition ending in a singleton); 800 are excluded by one game-free exact program each — levels, orderings, totals and the anchor conditions of Lemma 29 against $\rho > \frac{1}{2}$; 100 fall to a finite check with v free, both core memberships, the certificates of Theorem 23, the floor and ceiling stage-one pinnings (dualized as in Theorem 25), and the exact-cone facet inequalities as lazy cuts; and the remaining 108 fall to the same check with the complete multi-stage pinning ladders — every Lmin stratum pinned given the strata below it, every Lmax stratum pinned given the strata above it, the reduced-game caps encoded exactly as minima over preimages. All programs are infeasible in exact rational arithmetic with no indeterminate verdicts.

The second pipeline re-proves the same verdicts from a minimal hypothesis basis: singleton normalization and scale; core membership of both vectors; exactness imposed *existentially* (for each proper coalition S, an attaining core point w^S with $w^S(S) = v(S)$ — the definition of exactness, linear, with no appeal to the facet description); and Lmin/Lmax imposed through Lemma 31 and its mirror. All 2961 programs are again infeasible — so the theorems depend neither on Theorem 23 nor on the facet catalogue. The two pipelines agree configuration by configuration.

Soundness controls, both pipelines: dropping the pinning ladders (resp. certificates) restores feasibility wherever the configuration is possible at all; the hypothesis sets hold, in exact arithmetic, on 1050 profiled random exact games spanning 21 configurations — the first pipeline's 2874 stage certificates in one run, the second pipeline's 10,500 Farkas certificates in another, no failures in either; and on all eight configurations for which a game above some target ratio is known, the same programs run at that target return feasible models that *audit end-to-end* as genuine exact games in the assumed configuration — including the extremal configuration at ratio exactly $\frac{1}{2}$ and the $(4,1)$ regime at exactly $\frac{1}{3}$. The configurations declared possible include every configuration any harvest has produced. Attainment in Theorem 32 is by explicit games (111 harvested at exactly $\frac{1}{3}$). All drivers, logs and the coverage audit are archived with the verifiers. □

The two-regime picture is therefore a theorem: Lmin strata $(4,1)$ force $\rho \le \frac{1}{3}$ while $(3,2)$ allows $\frac{1}{2}$, both attained. One calibration deserves record: under the single-stage hypothesis set the final 108 configurations are all *satisfiable*, and every one of the 108 models audits as a genuine exact game — in the wrong configuration. The complete ladders are what make the encoding decisive rather than merely stronger; this is the fourth, and largest, instance of the certificate-not-realizable phenomenon in this program, and the reason every feasible model here is audited before it is believed.

6 Obstructions to an upper bound

6.1 The exact six-player bound

Conceptually, this is a theorem about a core polytope inside the normalized simplex. Singleton exactness makes that polytope touch every coordinate facet, through six witness allocations. If two canonical Lorenz selections were farther than 2/3, their coordinatewise comparison would have one of fifteen sign patterns. In each pattern, every infinitesimal exchange that improves one selection must be stopped by a tight coalition on which the exchange has negative sum; the opposite selection and the six facet witnesses impose seven necessary inequalities on that blocker. The finite part proves that no such blockers can coexist. Thus the computation is a fifteen-case convex-geometric exhaustion of necessary exchange obstructions, not an enumeration of games or core vertices.

The first nontrivial fixed dimension can now be closed without a generator normal form or a support-table census.

Proof architecture. There are three dependencies and only three. A human reduction maps any strict counterexample to one of fifteen sign-labelled systems of necessary exchange blockers. A finite linear-arithmetic lemma proves those systems infeasible, with independently checked proof objects. Finally, the explicit game $G(2,3)$ supplies the matching lower bound. The earlier generator census, equality census, continuity arguments, and support tables are historical discovery routes and are not used in this proof.

Theorem 34 (Six-player bound)**.** *Every singleton-normalized exact six-player game satisfies*

$$\|\operatorname{Lmin}(v) - \operatorname{Lmax}(v)\|_1 \le \frac{2}{3} v(N).$$

The bound is attained by $G(2,3)$, and hence $s(6) = 2/3$.

Proof. Scale $T = v(N)$ to one and write $x = \operatorname{Lmin}(v)$, $z = \operatorname{Lmax}(v)$. Put

$$P = \{i : x_i > z_i\}, \qquad M = \{i : z_i > x_i\}, \qquad E = \{i : x_i = z_i\}.$$

If $x \neq z$, both P and M are nonempty. Up to a common relabelling there are therefore fifteen triples (p, m, e) with $p, m \ge 1$ and $p + m + e = 6$, and

$$\|x - z\|_1 = 2 \sum_{i \in P} (x_i - z_i). \tag{28}$$

Exactness at each singleton supplies a core point q^i with $q^i_i = 0$. Three elementary selection anchors will be used. First, z cannot have a unique top player i: moving a short distance toward q^i would lower that unique maximum. Second, some z-top player has $x_i \ge z_i$, since otherwise a short move from z toward x lowers every old top coordinate. Dually, some x-bottom player has $z_i \le x_i$. Each move stays in the core by convexity and improves the relevant sorted lexicographic vector before a neighbouring stratum can cross.

Now let $h(N) = 0$. On the Lmin side call h admissible if its lowest affected payoff stratum has a positive coefficient and no negative coefficient. On the Lmax side call it admissible if its highest affected stratum has a negative coefficient and no positive coefficient. Such a direction strictly improves the relevant sorted lexicographic vector for a sufficiently short positive step. Optimality therefore forces a proper tight coalition S with $h(S) < 0$; otherwise all tight core inequalities have

nonnegative derivative and all other inequalities retain slack. Core membership of the opposite selection and the six witnesses gives, respectively,

$$z(S) \geq x(S), \quad q^i(S) \geq x(S) \ (i \in N), \tag{29}$$

or

$$x(S) \geq z(S), \quad q^i(S) \geq z(S) \ (i \in N). \tag{30}$$

As a finite differential control, the pivot implementation of admissibility was compared with a separately written first-affected-stratum criterion on all 4,683 labelled weak payoff orders, all 1,610 bound-two directions, and both selections: 15,079,260 comparisons with no mismatch.

For each sign triple, impose nonnegativity and efficiency of x, z, q^i, the strict signs, (28) with left side $> 2/3$, the three anchors, and the necessary blocker disjunction (29) or (30) for a finite menu of admissible integer directions. Thirteen cases use only efficiency-preserving directions with coefficients in $\{-1, 0, 1\}$; $(2, 3, 1)$ and $(3, 3, 0)$ use coefficients in $[-2, 2]$. After deletion of redundant direction clauses the fifteen menu sizes are

$$1, 1, 1, 1, 0, 2, 5, 14, 8, 2, 52, 71, 2, 90, 2, \tag{31}$$

in lexicographic order of (p, m, e). Every resulting exact QF_LRA formula is infeasible. Z3 and cvc5 agree on every formula; cvc5 emits CPC with proof pedantry enabled and trust disabled, and Ethos checks all fifteen proofs as `correct`. The formulas are relaxations: they retain only necessary consequences of tightness, so their infeasibility excludes every exact-game counterexample. The source-to-formula regeneration, hashes, proof controls, and all retained directions are in the repository under `verifiers/s6/strict_exchange_*`.

The two bound-two cases were additionally reconstructed by a separate encoder that imports none of the production encoder, lex predicate, blocker builder, direction generator, minimizer, or SMT serializer. Its base systems are satisfiable, its full systems are infeasible in both solvers, and its fresh trust-disabled CPC proofs check in Ethos. At the orbit level, the three largest menus admit a further unguarded reduction: their 213 labeled directions and 44 sign-role orbits shrink to 136 directions in 19 orbits. The $(3, 2, 1)$ and $(4, 2, 0)$ cases then use only one-for-one and two-for-two exchanges; the balanced $(3, 3, 0)$ case adds one asymmetric coefficient-two shape and its reverse. All three reduced formulas have separate trust-disabled CPC proofs checked by Ethos, and deleting any selected orbit makes the corresponding relaxation feasible. These controls reduce all three large cases using only the two cross-selection anchors; the no-unique-Lmax-top anchor is dispensable there. In $(4, 2, 0)$ they further reduce each blocker alternative to the opposite-allocation comparison and the four singleton-witness comparisons indexed by P; both M-indexed comparison families are dispensable, while each of the five retained sources is irredundant in the relaxation. Anchor localization is sharper: if the x-bottom and z-top anchors $a, b \in P$ coincided, then $x_m \geq x_a > z_a \geq z_m > x_m$ for $m \in M$. After relabelling the distinct anchors, a separate 18-real certificate uses nineteen directions in ten residual orbits and only the one exactness witness q^b. These controls reduce implementation correlation and make the finite anatomy readable. In that localized case two exchanges are automatic. Their negative blockers form a pair $\{b, 4\} \cup A, \{b, 5\} \cup B$ with $A, B \subseteq P \setminus \{b\}$. Of the 64 labelled pairs, 32 are base-infeasible and the other 32 form eleven classes under the residual symmetry; every class closes with a subset-minimal menu of at most seven remaining exchanges. Adding the two blocker inequalities also gives a short aggregate packet inequality by hand. This eleven-row decomposition has four hand-checkable rows. One is a two-exchange blocker relay whose four terminal blocker pairs force the distance back to at most $1/3$. In the second packet orbit the single exchange $-e_1 - e_3 + e_4 + e_5$ is forced to improve Lmin, while

every one of its twenty eligible blockers violates the opposite-allocation comparison. The third is a depth-one three-leaf star: the root activation, exact three-blocker list, and all child activations have structural inequalities. Its terminal Lmax leaf has sixteen possible complement cuts and its two terminal Lmin leaves reduce to four base cuts. Twenty explicit nonnegative weighted identities, checked by pure rational coefficient arithmetic without a solver, show that every such cut forces the distance to at most $1/3$. The fourth is a unique three-exchange chain. Its first twenty cuts reduce structurally to seven; six source-specific identities leave the unique first blocker, fifteen leave the unique second blocker, and sixteen exclude every terminal cut. The four hand-checkable rows cover fourteen labelled packets. More generally, five packet classes covering eighteen labelled packets close as forced-exchange relay trees; only the depth-three class remains machine-classified. The other six packet rows retain explicit open relay leaves. This does not replace the human obligation to verify the counterexample-to-formula implication.

Thus the upper bound holds. The family formula gives $\rho(G(2,3)) = 2(2(3-1)-1)/(3(2+1)) = 2/3$, proving equality. □

6.2 No constant below 2

Theorem 17 already rules out one shape of answer. Any bound of the form $\| \operatorname{Lmin}(v) - \operatorname{Lmax}(v) \|_1 \le c \cdot v(N)$ with a constant $c < 2$, valid for all n, is false: choose n even with $2 - 8/n > c$. So the only possible clean upper bound is dimension-dependent, and the right target is the finite-n statement

$$s_2(n) = 2 - \frac{8n}{n^2 - (n \bmod 2)} \qquad (n \ge 6),$$

now proved for the full exact class at $n = 6$ by Theorem 34, and confirmed by complete two-generator enumeration at $n = 7, 8$, together with the question of whether $s(n) = s_2(n)$ in larger dimensions.

6.3 No bound linear in the leximin floor

A second shape is also excluded, and this one was a live candidate. The inequality

$$\| \operatorname{Lmin}(v) - \operatorname{Lmax}(v) \|_1 \;\le\; 4(n-4)\,\mu(v), \qquad \mu(v) = \min_i \operatorname{Lmin}(v)_i,$$

holds with no violations in 900 fresh random five-player games and is tight both on the $n = 5$ extremal class and on $G(3,4)$ at $n = 8$, where both sides equal 12. It fails from $n = 9$ on:

game	n	$\| \operatorname{Lmin} - \operatorname{Lmax} \|_1$	$4(n-4)\mu$
$G(4,4)$	9	88/5	16
$G(3,5)$	9	33/2	15
$G(4,5)$	10	24	96/5
$G(5,5)$	11	95/3	70/3

The reason is structural rather than accidental: in $G(p,k)$ one has $\mu = p/(p+1) < 1$, bounded, while the gap grows like $2T$. No bound of the form $C(n)\mu(v)$ can track the divergence, whatever C is.

6.4 Non-monotonicity under majorization

The most natural route to an upper bound on the two-generator class is to argue that a uniform h-block is without loss of generality — the extremal family $G(p,k)$ has one — and then to optimize over g alone using the floor formula of Conjecture 8. That route is closed.

Define a *smoothing step* as a Robin Hood transfer of one unit from a larger to a smaller h-coordinate, so that h moves down in the majorization order, toward uniform. The premise that a smoothing step never decreases ρ is false, and not only for small blocks:

$\lvert B\rvert$	2	3	4	5
smoothing steps tested	40	76	84	84
violations	18	27	11	21
rate	45%	36%	13%	25%

In particular it fails for $|B| \geq 4$, so the block-size threshold one might guess from the $n = 5$ anomaly does not exist. An explicit verified counterexample at $|B| = 4$, with $n = 7$ and $T = 10$, both allocations confirmed in the core:

$$g = (0,5,0,3,2,0,0), \qquad h = (1,0,2,0,0,5,2),$$

$$\mathrm{Lmin} = (1,2,1,1,1,3,1), \qquad \mathrm{Lmax} = \left(\tfrac{5}{6}, \tfrac{5}{2}, \tfrac{5}{6}, \tfrac{4}{3}, \tfrac{7}{6}, \tfrac{5}{2}, \tfrac{5}{6}\right), \qquad \rho = \tfrac{1}{5}.$$

Smoothing the h-block $(5,2) \mapsto (4,3)$, i.e. replacing h by $(1,0,2,0,0,4,3)$, gives

$$\mathrm{Lmin} = \mathrm{Lmax} = (1,2,1,1,1,2,2), \qquad \rho = 0.$$

So a single Robin Hood step inside one generator collapses the ratio from $1/5$ to 0, and ρ is not monotone under majorization of h.

The two facts sit together uncomfortably and both are true: smoothing makes a game more symmetric and tends to *collapse* the two selections onto each other, yet the extremal family has a perfectly uniform h. The dependence is therefore genuinely non-monotone, and any upper-bound proof must handle general h directly rather than reducing to the uniform case.

7 What the results do and do not establish

Proved. Lemma 1 (lower-envelope generation and its converse); Lemma 3 (ceiling and overlap identity); Lemma 4 (support disjointness); Theorem 6 (two-generator core duality, with identity (1) valid without disjointness); Theorem 7 ($\mu \leq \lambda$); Proposition 9 (the cover inequality (2) for balanced covers); the validity half of Observation 10; Theorem 13 with Lemma 14 (closed forms for $G(p,k)$, including membership for all $p \geq 1$, $k \geq 2$ and the exact list of tight coalitions); Corollary 15 and Remark 16 (the deficit identity, twice); Theorem 17 ($s(n) \geq 2 - 8n/(n^2 - (n \bmod 2))$ and $\sup_n s(n) = 2$); Lemma 20 (core closing); Proposition 22 (the covering obstruction); Theorem 21 ($\rho = 1/2$ at $n = 5$, by hand). The failure of the $4(n-4)\mu$ bound from $n = 9$, and the non-monotonicity of ρ under majorization of h, are proved by the explicit games displayed in Section 6: each is a finite exact-rational computation on a named game, and each is a refutation, so a single instance suffices.

Proved by exhaustive finite check. Theorem 23 ($\mu = \lambda$ and $M = \nu$ for every exact five-player game; 1142 symmetry classes, each infeasible in exact rational arithmetic), whose M-side Corollary 27 consumes; and Theorems 33 and 32 ($s(5) = 1/2$ and the $(4, 1)$ bound $1/3$; 2961 configurations, two independent pipelines, machine-audited coverage); and Theorem 34 ($s(6) = 2/3$; fifteen table-free sign-count relaxations, independently checked exact CPC proofs). Theorem 25 is now a *hand* result (its one-page proof replaced the original 10,974-program check, which remains archived as confirmation); Lemma 29 and Corollary 30 are hand results; Lemma 31 is a hand result derived independently twice within the program. The checks are proof dependencies; the corollary's remaining additional hypothesis (the ν-attaining triple) is open in general and noted under (4) below.

Enumerated, not proved. The entries of Table 2 are maxima over the stated integer ranges, hence complete only for rational shapes whose primitive integer representative has total in that range; they are lower bounds for $s_2(n)$, not its value. The strict failure of equality in (3) is established by the displayed six-player game, but the claim that this is the *smallest* such failure rests on the enumeration. The tightness rates for (3) and for Conjecture 8 are sample statistics over 586 random games. The structural findings about the $n = 5$ extremal class in Section 5.3 — unique solution pair up to permutation, $(3, 2)$ and $(1, 4)$ stratum compositions, $|\operatorname{supp}(\mathrm{Lmax})| = 4$, the refutation of the seesaw hypothesis — are computational, over 360 profiled games; of these only the two-regime split is now proved (Theorem 32), the rest remaining sample statistics. The local-optimality certificates from seeded c-space climbing at $n = 5, 6, 7$ are numerical searches over the full exact class; the unseeded variant is under-powered at $n \geq 6$ and certifies nothing.

Conjectured. Four problems remain open.

(1) **Conjecture 8**, equality in the leximin floor formula. Reduced by linear-programming duality to the fractional-cover inequality (2); proved for balanced covers (Proposition 9); the over-covering case is open in general. At five players Theorem 23 *proves* the equality, for every exact game and not only two-generator ones. Verified in 586/586 random two-generator games at $n = 4, \ldots, 7$.

(2) **An upper bound for $n \geq 7$.** Theorem 17 forces any true bound to be asymptotically 2, so every bound $c \cdot v(N)$ with $c < 2$ is false for large n. The right target is $s_2(n) = 2 - 8n/(n^2 - (n \bmod 2))$ for $n \geq 7$, confirmed by complete enumeration at $n = 7, 8$ only. The six-player case is now proved: $s(6) = s_2(6) = 2/3$ (Theorem 34). Section 6 closes the majorization route and the $C(n)\mu$ route. $n = 5$ is settled (Theorem 33); larger fixed dimensions remain open.

(3) **Is the two-generator class extremal?** Adding a generator lowers the envelope and grows the core, so this is not automatic. All evidence is negative for a counterexample: zero improvements from about 200 third generators at each of $n = 5, 6, 7, 8$, and no improvement from seeded c-space climbing over the full exact class at $n = 5, 6, 7$. Open globally; settled affirmatively at $n = 5, 6$, where the two-generator lower bounds equal $s(5)$ and $s(6)$.

(4) **A structural explanation of the $n = 5$ versus $n \geq 6$ regime change.** The question now has a sharp two-part form, replacing the earlier "why five". *Part one, answered*: the starved family generalizing the extremal configuration (four players at $T/4$, a poor stratum of $n - 2$ absorbing the zeros) exists at *every* $n \geq 5$ with ratio exactly $(n - 4)/(n - 3)$ — now a theorem, by a six-row closed-form construction whose hand proof rests on two coalition values (the identity $T - (n - 3)\mu = 3T/4 = 3M$ lets one cap pin the poorer side from below and fix the ceiling from above, uniformly in n; archived with machine guards at $n = 5..9$).

At $n = 5$ the member realizes the extremal configuration and solution pair of Theorem 21 at ratio $\frac{1}{2}$ — though as a *second, non-isomorphic* game (no relabelling aligns the two; the extremal configuration is attained by at least two essentially different games) — so the family is tight there; its value *exceeds* the two-generator bound at $n = 5$, *ties* it at $n = 6$, and undershoots thereafter (it saturates at 1 while s_2 climbs to 2, so it is dominated for large n and says nothing about the supremum beyond $n = 5$); the regime change is the handover between champion families. The crossover test is now resolved by Theorem 34: both families attain the common value $s(6) = 2/3$. *Part two, open*: the floor side of Theorem 23 reduces (archived, with hand lemmas) to a single inequality — for every ordered pair (p, q) of players, some core point satisfies $2y_p + y_q \geq 3\lambda$ — needed only in the regime $\min_{S \subsetneq N} f(S)/|S| \leq T/5$ (outside it the equality $\mu = \lambda$ holds for every game with a nonempty core), and sharp: the margin vanishes on real games. Any proof must engage the dual region in which the efficiency multiplier is negative (active on about 1% of sampled instances), cannot factor through contraction to fewer players, and cannot proceed by saturating the nested pair of caps (both routes are refuted in the archive); and it must consume *exactness*: a balanced non-exact five-player game exists, in the relevant regime, with $\mu = \frac{1}{18} < \frac{1}{12} = \lambda$ (archived and independently re-verified), so core membership alone provably does not suffice. Remaining residues: the M-side of Theorem 23 by hand, a hand distillation of the 2961-configuration check of Theorem 33 (archived progress: 95.5% of configurations fall to twelve hand lemmas), and Corollary 27's ν-attaining-triple hypothesis.

Not claimed. Exactness theory, lower-envelope exactification, and the characterizations of the egalitarian selections are prior art (Arin and Iñarra, 2001; Arin et al., 2003, 2008; Biswas et al., 1999; Dietzenbacher, 2021; Dietzenbacher and Yanovskaya, 2020; Llerena and Mauri, 2017; Csóka et al., 2011; Studený and Kratochvíl, 2022); the four-player rigidity is the companion paper's classification (Flanagan, 2026) combined with the Lorenz implication of Llerena and Mauri (2017). The candidate new content is the two-generator core duality and its one-sided floor certificate, the family $G(p, k)$ with its closed forms and deficit identity, the resulting evaluation $\sup_n s(n) = 2$, the identification of the $n = 5$ anomaly with a proof by hand, the one-coalition certificate theorem and the pinned extremal configuration at five players, the evaluation $s(5) = \frac{1}{2}$ with the $(4, 1)$-regime bound $\frac{1}{3}$, and the two closed routes of Section 6, together with the direct exactness- exchange evaluation $s(6) = \frac{2}{3}$. Novelty remains conditional on specialist review.

Reproducibility note

Sections 2–4 and 6 are proved by hand and depend on no program. Theorems 23, 32, 33, and 34 are proved by exhaustive finite exact-rational checks, described in their proofs; those checks are proof dependencies, archived in `../verifiers` with the theorem-to-driver map in its README (for Theorems 32 and 33, both pipelines, the coverage audit, and the full control suite; for Theorem 34, the fifteen source-regenerated QF_LRA formulas, Z3/cvc5 records, trust-disabled CPC proofs, and Ethos checks under `s6/strict_exchange_*`, plus the independently formulated bound-two reconstruction under `s6/independent_bound_two/`). The literal 72-assertion human-to-machine map is audited under `s6/strict_exchange_formula_interface.json`. The orbit anatomy under `s6/strict_exchange_large_case_templates.json` is explanatory; the later 19-orbit formulas under `s6/orbit_core_certificates/` are separately proof-checked reduced alternatives for the three large cases. The anchor-localized $(4, 2, 0)$ alternative under `s6/p4_anchor_core_certificates/` is separately proof-checked and uses only one singleton exactness witness. Its explanatory eleven-row

forced-blocker decomposition is regenerated by `s6/audit_strict_exchange_p4_packet_classes.py`; its forced-exchange trees are regenerated by `s6/audit_strict_exchange_p4_relay_trees.py`. Theorem 25 is proved by hand; its superseded finite check and the hand proof's own symbolic guards are archived alongside. The remaining programs are bug audits, not proof dependencies. Exact rational programs in the companion repository independently reconstruct all displayed games and coalition values, the closed forms of Theorem 13 at twelve parameter pairs, the full coalition-inequality check and the tight-set prediction of Lemma 14 for every (p, k) with $1 \le p \le 7$, $2 \le k \le 7$ and $n \le 13$, and the four-way agreement on the game of Theorem 21. They are bug audits, not proof dependencies. All arithmetic is exact rational; no floating-point value appears in any stated result. Reported optima come from a from-scratch exact-rational two-phase simplex with Bland's rule, from a staged exact solver over linear real arithmetic, or from exact vertex enumeration, with at least two methods agreeing on every displayed allocation; one library simplex implementation cycles on these degenerate programs and was not used for any reported number.

References

Arin J, Iñarra E (2001) Egalitarian solutions in the core. International Journal of Game Theory 30:187–193.

Arin J, Kuipers J, Vermeulen D (2003) Some characterizations of egalitarian solutions on classes of TU-games. Mathematical Social Sciences 46:327–345. https://doi.org/10.1016/S0165-4896(03)00051-9

Arin J, Kuipers J, Vermeulen D (2008) An axiomatic approach to egalitarianism in TU-games. International Journal of Game Theory 37:565–580. https://doi.org/10.1007/s00182-008-0133-6

Biswas AK, Parthasarathy T, Potters JAM, Voorneveld M (1999) Large cores and exactness. Games and Economic Behavior 28:1–12. https://doi.org/10.1006/game.1998.0686

Csóka P, Herings PJJ, Kóczy LÁ (2011) Balancedness conditions for exact games. Mathematical Methods of Operations Research 74:41–52.

Fujishige S (1980) Lexicographically optimal base of a polymatroid with respect to a weight vector. Mathematics of Operations Research 5:186–196.

Gangam RR, Garg N, Shahkar P, Vazirani VV (2024) Equitable core imputations for max-flow, MST and b-matching games. Preprint, arXiv:2403.06037.

Klijn F, Vermeulen D, Hamers H, Solymosi T, Tijs S, Villar JP (2003) Neighbor games and the leximax solution. Mathematical Methods of Operations Research 58:191–208.

Llerena F, Mauri L (2016) Reduced games and egalitarian solutions. International Journal of Game Theory 45:1053–1069.

Vazirani VV (2024) Fair core imputations for the assignment game: new solution concepts and efficient algorithms. Preprint, arXiv:2402.11437.

Dutta B, Ray D (1989) A concept of egalitarianism under participation constraints. Econometrica 57:615–635.

Dietzenbacher B (2021) Monotonicity and egalitarianism. Games and Economic Behavior 127:194–205. https://doi.org/10.1016/j.geb.2021.03.006

Dietzenbacher B, Yanovskaya E (2020) Antiduality in exact partition games. Mathematical Social Sciences 108:116–121. https://doi.org/10.1016/j.mathsocsci.2020.04.006

Flanagan M (2026) Balanced transport at the egalitarian core: a complete boundary phase diagram. Preprint; companion paper.

Llerena F, Mauri L (2017) On the existence of the Dutta–Ray's egalitarian solution. Mathematical Social Sciences 89:92–99. https://doi.org/10.1016/j.mathsocsci.2017.07.002

Studený M, Kratochvíl V (2022) Facets of the cone of exact games. Mathematical Methods of Operations Research 95:35–80. https://doi.org/10.1007/s00186-022-00770-4